

Nano-technology enhanced biosensor for early bacterial sepsis detection / Sepsis biomarker analysis using SERS based sensor

Masters Project Report – 3rd Semester

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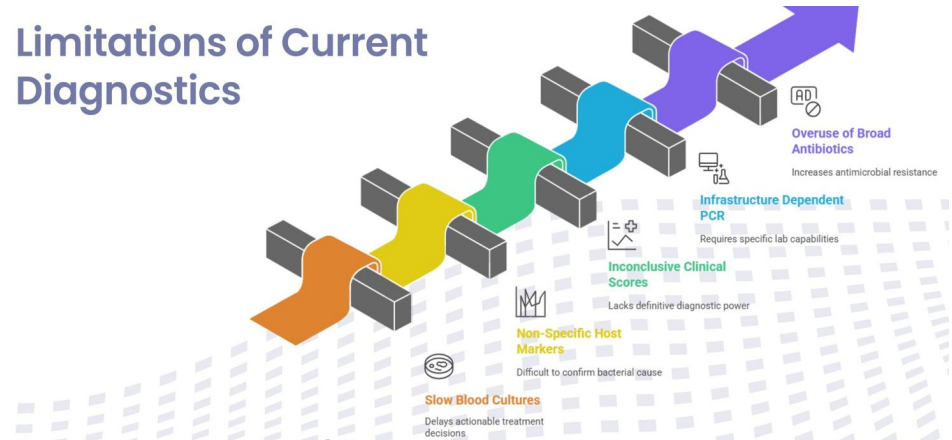
Clinical Urgency of Sepsis

- 49M cases, 11M deaths annually
- 85% of deaths in low-resource settings
- Delayed diagnosis increases mortality
- Empirical antibiotics prescribed without pathogen knowledge
- Contributing to antimicrobial resistance crisis
- Critical Need: Rapid, bacterial-specific point-of-care diagnostic

Diagnostic Gap - Problem Statement

- Blood Culture: 24–72 hrs, bacterial-specific, slow
- PCT/CRP Biomarkers: 1–2 hrs, non-specific
- PCR Methods: 1–6 hrs, target-dependent, expensive
- SERS Solution: <30 min, bacterial-specific, novel approach

Limitations of Current Diagnostics



Current Diagnostic Methods & Limitations

- Blood culture delays ? antibiotic overuse
- Biomarkers cannot distinguish bacterial vs viral/fungal
- PCR requires pre-selected targets, expensive
- Conclusion: Need for novel diagnostic approach

Why Bacterial-Specific Biomarkers?

- Early diagnosis and improved clinical outcomes.
- Directly indicate the presence of bacteria (not just host inflammation).
- Enable Gram classification. Reduce unnecessary broad-spectrum antibiotic use and combat AMR.



Traditional Approach

Host response detection
Delayed immune activation
Non-specific inflammation
Cannot guide antibiotics
Incremental improvement



Bacterial Specific Marker

Direct bacterial detection
Immediate bacterial identification
Bacterial specific marker
Enables targeted therapy
Paradigm shift

Background

Difference	Gram-Positive	Gram-Negative	Clinical Significance	Reference
Cell Wall Structure	Thick peptidoglycan (20-80 nm)	Thin peptidoglycan + outer membrane	Determines antibiotic penetration and SERS targeting strategy	Silhavy et al. Cold Spring Harb Perspect Biol 2010;2(5):a000414
Peptidoglycan Content	90% of cell wall mass	5-10% of cell wall mass	Major structural target for β -lactam antibiotics and detection	Vollmer et al. FEMS Microbiol Rev 2008;32(2):149-167
Outer Membrane	Absent	Present (barrier function)	Primary resistance mechanism; barrier to drugs and immune system	Nikaido H. Microbiol Mol Biol Rev 2003;67(4):593-656
Lipopolysaccharide (LPS)	Absent	Present (endotoxin component)	Major sepsis mediator; triggers cytokine storm and shock	Raetz & Whitfield. Annu Rev Biochem 2002;71:635-700
Gram Staining	Purple/Violet (retains stain)	Pink/Red (loses stain)	Primary laboratory identification method (95% accuracy)	Gram HC. Fortschr Med 1884;2:185-189
Antibiotic Susceptibility	Generally, more susceptible	More resistant (outer membrane)	Guides empirical antibiotic selection in sepsis management	Delcour AH. Biochim Biophys Acta 2009;1794(5):808-816
Endotoxin Production	No endotoxin	Lipid A endotoxin present	Key pathophysiology difference in sepsis severity and outcomes	Rietschel et al. FASEB J 1994;8(2):217-225
Clinical Presentation	Device infections, pneumonia	UTI, septic shock, HAI	Influences infection source, severity, and treatment approach	Martin et al. N Engl J Med 2003;348(16):1546-1554

Emerging Biosensor Technologies

- Electrochemical biosensors: rapid but limited validation
- Optical biosensors (SPR): label-free but complex instrumentation
- SERS

Why SERS? – Scientific Rationale

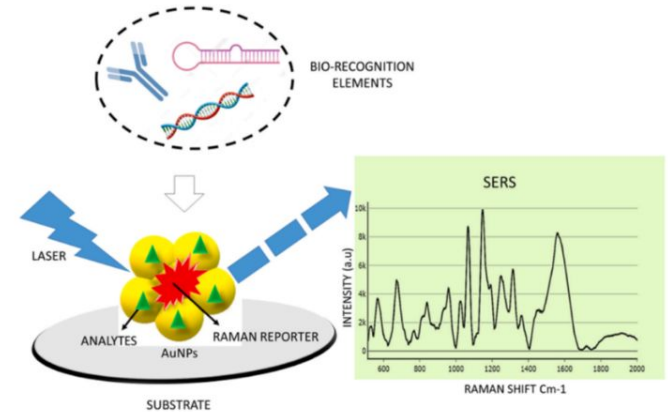
- Enhancement factor: 10^6 – 10^{14} ; $\sim 100\times$ better than alternatives
- Plasmonic enhancement mechanism amplifies Raman signals
- Molecular fingerprinting without labels or amplification
- Clinical feasibility with Au–Ag nanostructures
- Literature support: publications validating SERS

SERS Nanobiosensor: Overview

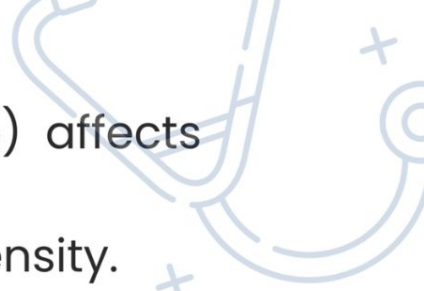
- SERS = Surface-Enhanced Raman Spectroscopy; provides molecular fingerprints.
- Plasmonic nanostructures (Ag/Au) create strong local fields (hotspots).
- Combine SERS substrate with capture probes (aptamers/antibodies) for selectivity.

SERS Fundamentals

- Raman scattering measures vibrational modes; specific peaks = molecule signature.
- Electromagnetic enhancement from plasmon resonance scales..
- Hotspots (nanogaps) give the largest enhancement; single-molecule sensitivity possible.

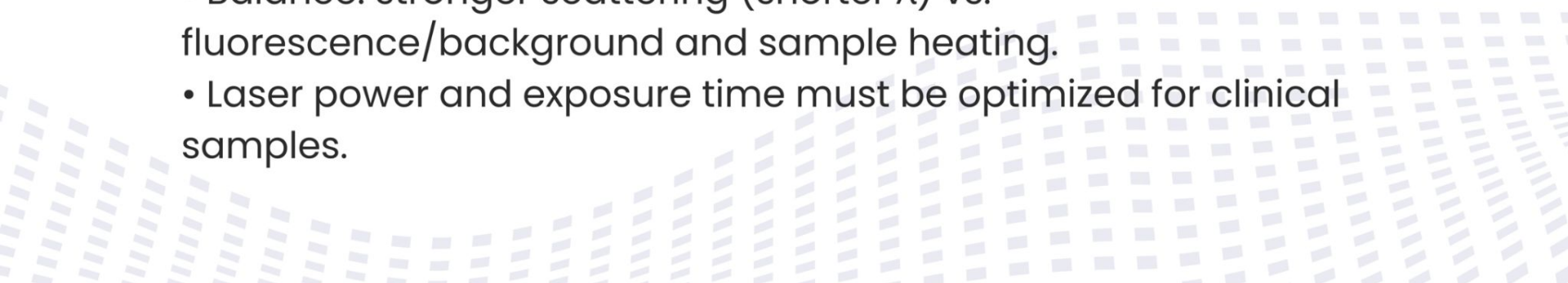


Nanomaterials & Substrate Design



- Choice of metal (Ag stronger SERS; Au more stable) affects resonance.
- Particle geometry (stars, rods, gaps) tunes hotspot intensity.
- Substrate uniformity & reproducible fabrication are essential for clinical use.

Excitation Wavelength & Practical Considerations

- Common lasers: 532 nm (resonant dyes), 633 nm, 785 nm (bio-friendly), 1064 nm (high autofluorescence samples).
 - Balance: stronger scattering (shorter λ) vs. fluorescence/background and sample heating.
 - Laser power and exposure time must be optimized for clinical samples.
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Project Objectives

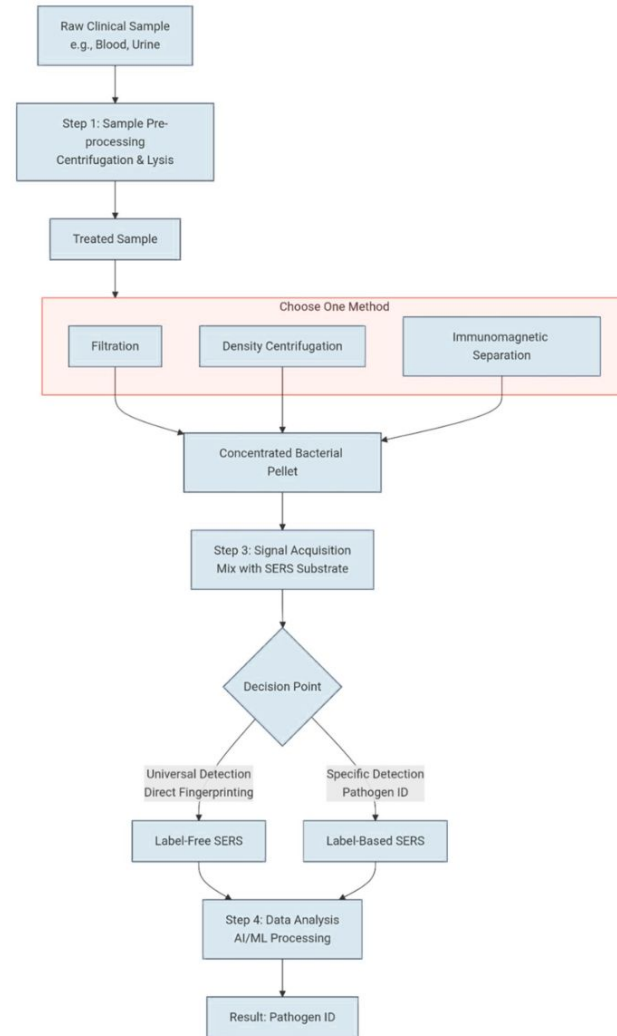
- ★ Design Ag/Au bimetallic SERS substrates (target EF > 10^8 ; size 10–100 nm)
- ★ Identify Raman signatures: PGN (universal), LPS (Gram–), LTA (Gram+)
- ★ Rapid, label-free detection: 50–100 μ L plasma; femtomolar sensitivity; total workflow 25–35 minutes
- ★ POC feasibility: portable instrumentation; <30 min turnaround; USD 2–5 per test; automated analysis

Methodology - Workflow

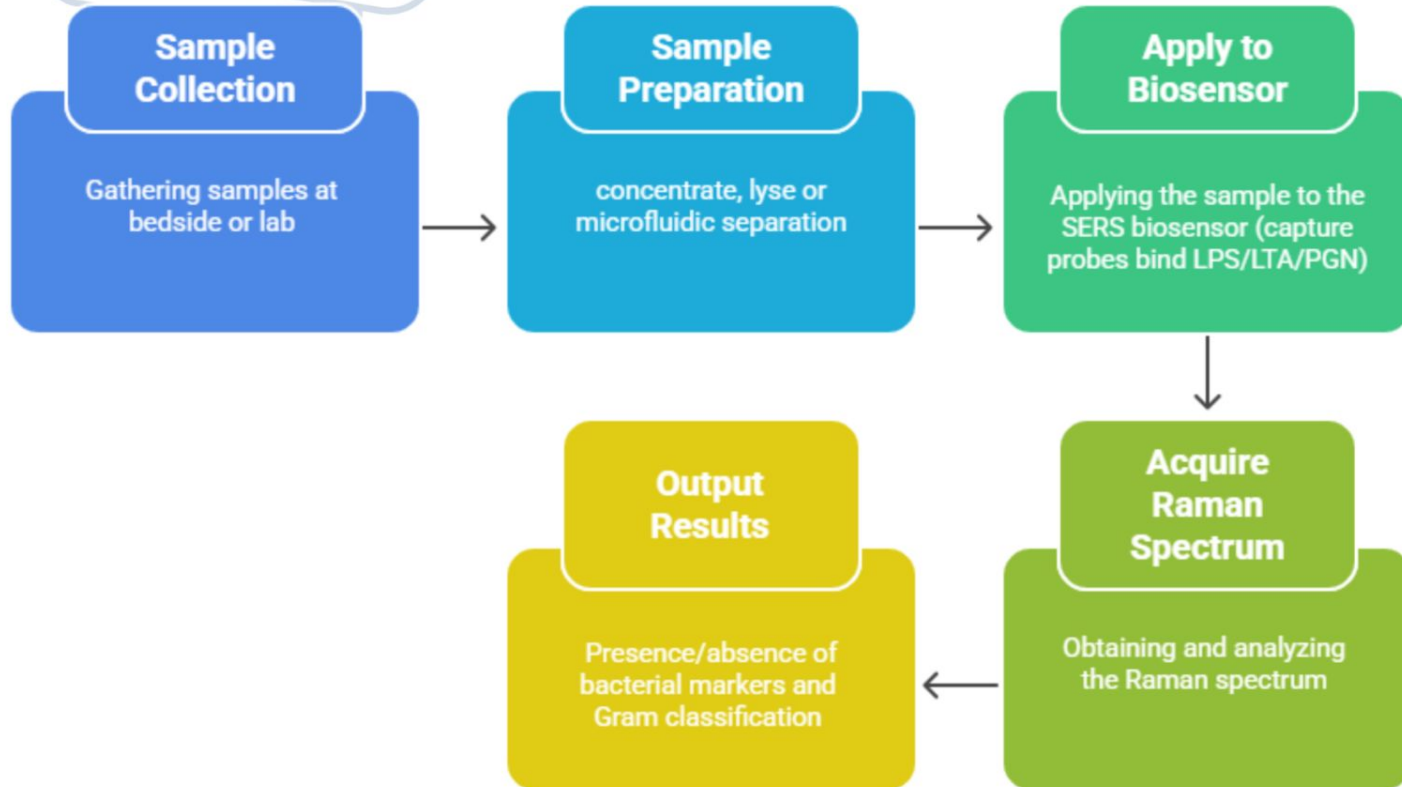
- Literature review and biomarker selection
- Fabrication of SERS substrates on silicon with Ag/Au nanostructures
- Standardization of sample deposition techniques
- Raman spectral acquisition using bench-top Raman system
- Preprocessing: baseline correction, normalization, noise filtering
- Feature extraction and peak assignment
- Machine learning–assisted spectral classification (future phase)
- Comparative analysis with traditional biomarker assays (planned)



Workflow Pipeline:



Diagnostic Workflow (Patient → Answer)



Work completed

Literature Survey on Bacterial-Specific Biomarkers for SERS-Based Sepsis Detection:

- Conducted comprehensive systematic review of research publications , analyzing bacterial-specific surface biomarkers with established Raman spectroscopic signatures and demonstrated SERS detectability. Critical evaluation of biomarkers superior to traditional host-response markers (PCT, CRP) for enabling rapid, bacterial-specific sepsis diagnosis.

Collaborative Review Meetings with Principal Investigators and Research Team:

- Conducted multiple strategic review meetings with project supervisors and research collaborators to discuss preliminary findings, obtain expert input on biomarker selection rationale, and identify optimization priorities.
- Meetings resulted in: Confirmation of biomarker panel selection aligning with clinical sepsis diagnostic needs. Molecular simulations suggestions with optimization and frequency analysis and strategic planning for next phase approaches.

Finalization of Bacterial Biomarker Panel:

Focus Area

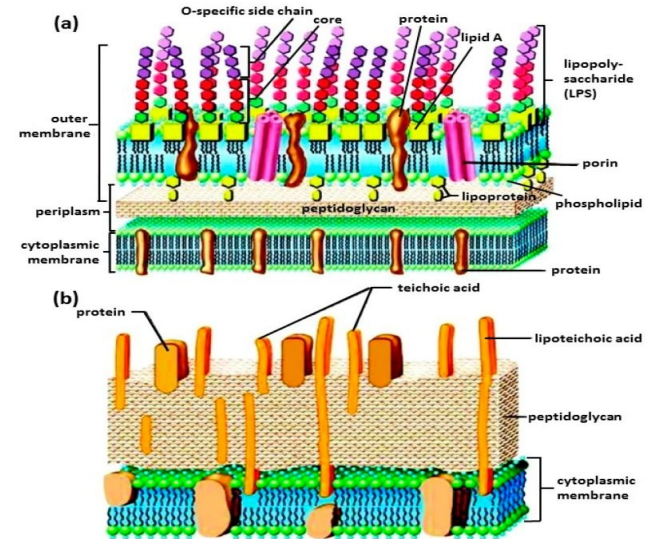
- Markers specific to the surface of bacteria and not overlapping with other pathogens / Readily available or expressible markers only in bacteria.
- Help healthcare professionals with confirmation of a bacterial infection and subsequent gram classification for early diagnosis and treatment and SERS based detection compatibility.

Rationale for biomarkers selection:

1.Universal Bacterial Detection :N-acetylmuramic acid

Exclusivity:

- Present in ALL bacteria, absent in viruses/fungi/parasites
- Direct bacterial confirmation
- Unique to bacterial peptidoglycan only



2. Gram classification

2.a **Gram-Positive–Specific Markers** : Lipoteichoic Acid (LTA)

Uniqueness:

- Only in Gram-positive bacteria , surface accessibility .

2b. **Gram Negative specific marker** :Lipopolysaccharides (Lipid A , O antigen)

Uniqueness:

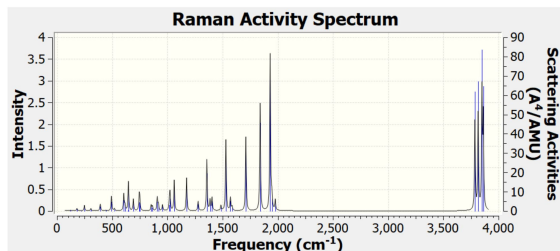
- Only in Gram-negative bacteria, absent in all other organisms
- Strong enhancement, endotoxin correlation .

Established optimized three-component biomarker panel: (1)N-acetylmuramic acid (PGN) - universal bacterial detection marker present in all (2) Lipopolysaccharide (LPS) - Gram-negative specific marker (3) Lipoteichoic Acid (LTA) - Gram-positive specific marker.

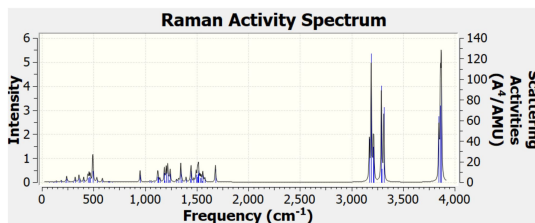
- Panel selected based on biomarker exclusivity to bacteria, surface accessibility for SERS detection, and Gram classification capability.
- These markers can be supplemented with bacterial-specific components to enhance detection sensitivity in complex cases where signals might be obscured.

Optimization and Preliminary SERS Characterization with Test Molecules:

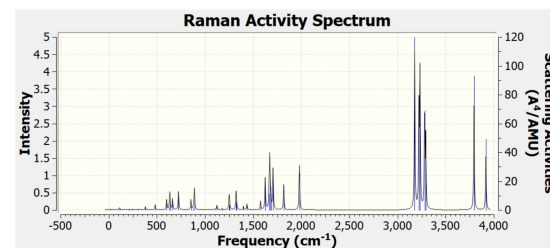
- Initiated experimental optimization of SERS and Raman Spectral Acquisition and Preliminary Preprocessing. Reviewing of Raman Peaks Assignment and Biomarker Signature Identification for reliable biomarker discrimination using software simulation
- Simulations pertaining to Raman optimization and frequency using basic molecules using Gaussian.



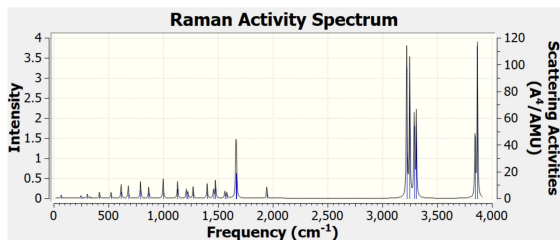
Uric Acid



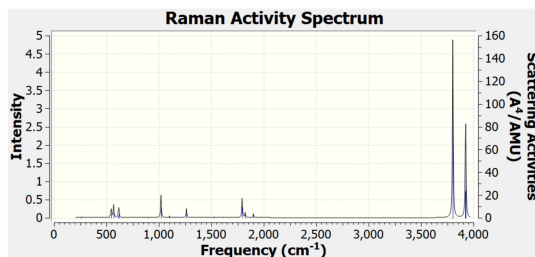
Glucose



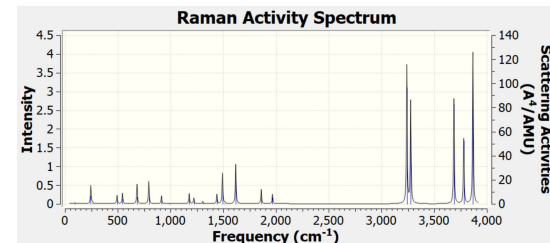
Creatine



Lactic acid



Urea



Glycine

Groundwork for Interpretation and analysis of raman spectrum using libraries and comparative analysis with literature along with preliminary wavelength assessment review

Library-Based Matching:

- Using open molecular Raman libraries (e.g., **Raman Open Database**, **RamanKit**, **SpectraBase**).
- Match measured peaks with standard fingerprint regions for biomolecules (proteins, lipids, bacterial cell-wall components).

Comparative Literature Analysis:

Verification of identified peaks with published spectra for sepsis biomarkers (PGN, LPS, LTA, CRP, IL-6, etc.).

Outcome

A structured base to validate SERS spectral peaks against known molecular vibrational signatures.

Preliminary Wavelength Assessment & Replication of Literature Spectrum

- We are currently replicating the Raman acquisition protocol used in the selected reference study.
- Preliminary measurements at 785 nm are ongoing, and our goal is to achieve comparable peak positions and intensities for validating our SERS substrate performance.

Initiation of Pure Biomarker Sample Procurement Procedures

- Initiation of procurement procedures for commercially available high-purity bacterial biomarker standards: Trying to establish purchase with quality control specifications ensuring received biomarkers meet analytical standards for SERS characterization and clinical validation studies.

Planning and Strategy for Next Phase of Work

- Developed comprehensive roadmap for subsequent project phases: completion of SERS substrate optimization with pure biomarker standards, establishment of calibration curves, reproducibility assessment across multiple substrate batches; - prospective validation with clinical samples, exploring machine learning model development.
- Exploring Feasibility of Portable, Low-Cost POC Raman Device suitable for clinical deployment. automated spectral analysis algorithms for rapid biomarker identification without requiring specialized operator training; integration into emergency department and ICU workflows enabling true point-of-care deployment.

Novelty

- ❖ No identical patent
- ❖ Novel use of PGN + LPS + LTA with SERS
- ❖ Integrated POC SERS workflow = patentable

Competitive Advantage & Market Differentiation

Feature	Blood Culture	PCT/CRP Tests	SERS Biosensor
Time-to-Result	24-72 hours	~1 hour	< 30 MINUTES *
Pathogen Specificity	Yes	No	YES (Bacterial & Gram-Type)
AMR Guidance	Delayed	No	IMMEDIATE
Platform	Central Lab	Point-of-Care	POINT-OF-CARE

Impact

Clinical & Societal Impact

- Early targeted therapy = reduced mortality
- Narrow-spectrum antibiotics reduce AMR
- Affordable point-of-care solution and healthcare savings

Market opportunity

- Sepsis diagnostics: \$1.14B ? \$2.5B (2030)
- SERS biosensors: \$412M ? \$1.38B (2033)
- High revenue potential

Challenges

- Reproducibility of SERS substrates across batches with Batch-to-batch nanoparticle variation affecting enhancement factors
- Matrix Interference: Blood components cause non-specific signals
- Biological variability in plasma samples and Noise/interference in real patient samples
- Sensitivity Achievement: Detecting clinically relevant biomarker concentrations requires optimization
- Integration of Raman hardware into a compact POC system
- Need for large-scale clinical validation
- Regulatory approval challenges for medical diagnostics and devices
- Clinical Adoption: Overcoming clinical practice inertia and technology skepticism

Conclusion

- SERS nano-biosensor offers rapid, bacterial-specific sepsis diagnostics
- Novel integrated workflow with Gram classification
- Potential to transform sepsis management globally.

This preliminary project phase has successfully established the scientific and technical foundation for developing a SERS-based nano-biosensor platform for rapid bacterial-specific sepsis diagnostics.

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**Thank
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